

Quantum Tangent Kernel

Milestone 2

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What is a kernel function?

- Consider 2 spaces of dimensions m and n . ($m \gg n$)
- We have 2 vectors $X, Y \in R^n$ and we want to compute their dot product in R^m .
- Problem:
 - We don't know the mapping $\phi : R^n \rightarrow R^m$ and computing it could be computationally too expensive.

What is a kernel function?

- Suppose $n = 2$. i.e. $X = [x_1, x_2]^T$ and $Y = [y_1, y_2]^T$.
- Let us use the kernel function $k : R^2 \times R^2 \rightarrow R$
- $$k(X, Y) = (1 + X^T Y)^2$$
- Expanding:
 - $k(X, Y) = (1 + x_1 y_1 + x_2 y_2)^2$
 - $$= (1 + x_1^2 y_1^2 + x_2^2 y_2^2 + 2x_1 y_1 + 2x_2 y_2 + 2x_1 y_1 x_2 y_2)$$
- This is the dot product between the following vectors:
 - $X' = [1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1 x_2]^T$ and
 - $Y' = [1, y_1^2, y_2^2, \sqrt{2}y_1, \sqrt{2}y_2, \sqrt{2}y_1 y_2]^T$

What is a kernel function?

- By applying the kernel $k(X, Y) = (1 + X^T Y)^2$ on $X, Y \in R^2$,
 - We managed to find their dot product in R^6
 - Without explicitly computing $\phi : R^2 \rightarrow R^6$
 - $\phi(X = [x_1, x_2]^T) = [1, x_1^2, x_2^2, \sqrt{2}x_1, \sqrt{2}x_2, \sqrt{2}x_1x_2]^T$
- This is often called as the kernel “trick” because it allows operations in high dimensional spaces without actually doing computations in those spaces.
- Kernel functions are also called as feature maps because they implicitly compute the dot product in a higher-dimensional space without requiring explicit transformation.

Neural Tangent Kernel

- NTK is defined as:

- $$\Theta(x, x') = \sum_{i=1}^m \nabla_{\theta_i} f_{\theta}(x) \cdot \nabla_{\theta_i} f_{\theta}(x')$$

- $f_{\theta}(x)$ is the output of the network given input x .
- $\nabla_{\theta} f_{\theta}(x)$ is the gradient of the output with respect to parameters θ .
- NTK is computed by summing the inner products of gradients over all parameters in the network.

Neural Tangent Kernel

- NTK is defined as:

- $$\Theta(x, x') = \sum_{i=1}^m \nabla_{\theta_i} f_{\theta}(x) \cdot \nabla_{\theta_i} f_{\theta}(x')$$

- The NTK matrix $\Theta(x, x')$ quantifies how the network's prediction on x is influenced by changes in θ that also affect the prediction on x' .
- Large NTK values between two samples indicate that gradient updates from one sample will strongly influence the model's output on the other sample.
- Thus, NTK captures the implicit similarity between data points from the perspective of how the network learns.